

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import networkx as nx

import torch
import torch.nn as nn
import torch.optim as optim

from torch_geometric.data import Data
from torch_geometric.nn import GCNConv
```

✓ Homework 4

Networks of multivariate financial data and graph-based forecasting

We look at an example of multivariate daily futures data across various asset classes such as equity index, currency, fixed-income and commodity futures.

Note that '6B' and '6E' (GBPUSD and EURUSD), are renamed 'B6' and 'E6'. Everything else is as usual, e.g. ES (S&P 500), NQ (Nasdaq), etc.

Step 1. Get dataset

```
# Read your merged.csv
data = pd.read_csv("./Merged_Multivariate.csv")

data.drop("flag", axis=1, inplace=True) # remove "flag" column
data['date'] = pd.to_datetime(data['date'])
data.head()
```



	date	open	high	low	close	volume	vol	symbol
0	2008-01-02	1.9864	1.9879	1.9784	1.9817	47489	0.002877	B6
1	2008-01-03	1.9831	1.9854	1.9719	1.9748	62864	0.004098	B6
2	2008-01-04	1.9738	1.9866	1.9688	1.9740	74797	0.005405	B6
3	2008-01-07	1.9745	1.9771	1.9665	1.9709	44803	0.003229	B6
4	2008-01-08	1.9696	1.9858	1.9678	1.9724	50896	0.005469	B6

Note that the data has been supplemented with a new column called 'vol' which contains the Parkinson volatility computed from the daily high and low prices.

Exercise 1

Pivot the data to a wide table of open or close price with each column being an asset. Compute and visualise the correlation matrix. Are there any hierarchical structures?

Exercise 2

Construct and visualise various networks obtained from the data by thresholding the absolute value of the correlation. You can use the networkx library as imported above.

Challenge

Use a network constructed in Exercise 2 (you might want to ensure that this was done on the train data set) and build (a toy model) of a Graph Convolutional Network to forecast daily (Parkinson) volatility. You can use the torch_geometric library as imported above.

